Results

Task	Quantity	Your value
P1	rows, stores, weeks, holiday store-weeks	6,435 rows; 45 stores; 143 weeks; 450 holiday store-weeks
P2	$P(S)$, $P(H)$, $P(Q)$, $P(C)$, $P(F)$, $P(J)$, $P(G)$	$P(S) = 0.1265$; $P(H) = 0.0699$; $P(Q) = 0.1259$; $P(C) = 0.1537$; $P(F) = 0.4707$; $P(J) = 0.4696$; $P(G) = 0.4889$
P3	$P(S \cap H)$, $P(S \cap Q)$, $P(H \cap Q)$, $P(S \cap C)$	0.0190 (122); 0.0517 (333); 0.0280 (180); 0.0287 (185)
B1	$P(S \mid H)$, $P(S \mid H^c)$	0.2711 (122/450); 0.1156 (692/5,985)
B2	$P(H \mid S)$ by Bayes, and by counting	0.1499 by Bayes; 0.1499 by counting (122/814)
B3	holiday with the highest $P(S \mid g)$; group with the largest $P(g \mid S)$	Thanksgiving, $P(S \mid g) = 0.8222$; No holiday, $P(g \mid S) = 0.8501$
B4	$P(S \mid Q)$, $P(Q \mid S)$	0.4111 (333/810); 0.4091 (333/814)
B5	holiday rule: precision, recall; Nov–Dec rule: precision, recall	Holiday rule: precision 0.2711, recall 0.1499; Nov–Dec rule: precision 0.4111, recall 0.4091
B6	precision of the holiday rule if strong weeks doubled, if halved	Doubled ($P(S) = 0.2530$): 0.4652; halved ($P(S) = 0.0632$): 0.1478
N3	$P(S \mid$ profile a), $P(S \mid$ profile b)	0.7939; 0.0301
N6	accuracy, precision, recall: Naive Bayes / "never strong" / "strong if holiday"	Naive Bayes: 0.8682 / 0.0000 / 0.0000; Never strong: 0.8682 / 0.0000* / 0.0000; Strong if holiday: 0.8455 / 0.2556 / 0.0902

* "Never strong" flags no weeks, so its precision is undefined (0/0) and is reported as 0 (`zero_division=0`). Naive Bayes at the 0.5 threshold also flags no weeks, so its row is identical. Part B uses store averages from all 143 weeks; N3 and N6 use averages from the training years only (R3), with $P(S) = 0.1276$ (574/4,500) in training and $P(S) = 0.1318$ (255/1,935) in test.

Github: https://github.com/AchintyaShah25/walmart-naive-bayes-timespli.git